

DeepSPE: Predicting the Association between Photospheric Magnetic Field Parameters and Solar Proton Events Using Ensemble Deep Learning

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Abstract—In this work, we present an ensemble deep learning framework, named DeepSPE, which uses sampled photospheric magnetic field parameters or features from solar active regions (ARs), to predict the association between the features and a special kind of solar radiation known as solar proton events (SPEs). The association prediction problem is motivated by the fact that magnetic field parameters from ARs can provide important information for SPEs that could possibly occur within the ARs. Each sample or record of magnetic field parameters is labeled positive, meaning that the magnetic parameters are associated with an SPE, or negative, meaning that the magnetic parameters are not associated with an SPE. DeepSPE rewards features that act as better predictors of the class labels assigned to particular samples, where the chance of an identical selection of poor features throughout the ensemble is reduced, if not eliminated. With balanced class weights assigned to its loss function, DeepSPE is able to mitigate the class imbalance problem, whereby negative samples vastly outnumber positive samples, and provide valuable predictions of AR activity that could lead to the release of SPEs. Extensive experimentation demonstrates the good performance of the proposed framework and its suitability for being used to predict the association between measurements of solar activity and SPE occurrence.

Index Terms—Neural networks, solar radiation, space weather, information integration

I. INTRODUCTION

In an era where humankind has expanded its ambition for space exploration and relies increasingly on wireless communication and fine electronics, systems of predicting and mitigating the effects of harmful solar radiation are extremely valuable. A common strategy among developers of such systems is to use neural networks to analyze varying datasets containing various measurements of solar activity and associate solar activity with releases of harmful radiation in order to predict whether future activity could also result in harmful radiation being released. However, the use of single neural network estimators without any form of voting in an ensemble could cause models to fail even if data collected by them could otherwise be valuable. Common issues that any solution to this problem would need to address include class imbalance, the focus on specific kinds of event to monitor, and the value that any prediction may have, which may come not only from the accuracy or consistency of a model used to produce it, but also from any ability to know precisely when a release of harmful solar radiation may take place.

Solar flares are large bursts of radiation released into space; they appear as sudden and unexpected brightenings in the solar atmosphere with a duration ranging from minutes to hours. Coronal mass ejections (CMEs) are significant discharges of plasma and magnetic fields produced by the solar corona into the interplanetary medium. They are considered to be the largest-scale solar eruptions in the solar system and occur on a quasi-regular basis. Research shows that large flares and accompanied CMEs cause solar energetic particles (SEPs). SEPs, composed of electrons, protons, and heavy ions, are accelerated by magnetic reconnections and shock waves associated with CMEs.

An SEP event occurs when a burst of these energetic particles is released from the Sun. Among SEP events, the solar proton event (SPE) is of particular interest. The SPE is a type of SEP event, focusing specifically on the high-energy protons emitted from the Sun. When SPEs are strong, they cause nuclear cascades in the Earth’s upper atmosphere and also pose radiation hazards to equipment in space.

Solar active regions (ARs) manifest complex magnetic geometry and properties. ARs are the sources of flares and CMEs. The durations of ARs range from days to months. Recently, researchers combined machine learning (ML) with physical parameters derived from photospheric vector magnetograms provided by the Helioseismic and Magnetic Imager (HMI) [1] on board the Solar Dynamics Observatory (SDO) [2] to predict flares, CMEs, and SEPs. These physical parameters, including magnetic helicity and magnetic flux, are part of the photospheric vector magnetic data product, named the Space-weather HMI Active Region Patches (SHARPs) [3], produced by the SDO/HMI team. The SHARP magnetic field parameters are used for Solar Cycles 24 and 25. The SDO/HMI team also created a data product, called the Space-Weather MDI Active Region Patches (SMARPs) [4], derived from maps of the photospheric magnetic field taken by the Michelson Doppler Imager (MDI) [5] on board the Solar and Heliospheric Observatory (SOHO) [6]. The SMARP magnetic field parameters are used for Solar Cycle 23. Kasapis et al. [7] integrate the SHARP and SMARP magnetic field parameters into a single dataset, hereafter referred to as SHMARP. The SHMARP magnetic field parameters can be used in Solar Cycles 23, 24, and 25.

In this paper, we present an ensemble deep learning framework, named DeepSPE, to predict the association between SHMARP magnetic field parameters and SPEs. In the next section, we review previous studies that are closely related to ours.

II. RELATED WORK

A. Ensemble Deep Learning in SPE Prediction

The concept of ensemble deep learning, that is, the combination of the predictions of multiple deep learning estimators into single predictions, in SPE prediction, is not unfounded. In particular, Aminalragia-Giamini et al. [8] describes a way in which it can be used, specifically, by using a committee of single neural network (SiNN) estimators that use all of the available numerical features from a dataset of solar flare measurements. SiNNs may end up with poorly optimized parameters, even with good data, and could make poor predictions. Ensemble learning would allow for more chances to succeed in making predictions, even if some estimators do poor jobs at doing so, and for more flexibility in random sampling and sample counts, with suboptimal setups more likely to occur when there are fewer samples or when a particular randomly chosen subsample may not be representative of a general population at large. This would also allow for greater flexibility in updating models, since new estimators can be added and unneeded ones can be removed from an ensemble without needing to retrain all estimators in it.

B. The Random Hivemind Algorithm

While the ensemble deep learning approach mentioned above is promising, an unfortunate twist of fate could still mean that an entire ensemble of neural networks could all end their trainings with the same bad final set of weights and biases. To remedy this issue, O’Keefe et al. [9] propose a solution known as the Random Hivemind (RH) algorithm, where features of a dataset of solar flare measurements are randomly bagged in limited quantities to be chosen as estimators for given ensembles. The features are chosen in a “weighted roulette wheel” format: they are scored based on importance in making predictions and the chances they have for being selected are reduced or increased based on these scores. An estimator’s epoch count and learning rate are also boosted or attenuated based on the importance scores of the features that are selected.

C. Active Region-Based SPE Prediction

The two studies mentioned above make use of flare databases for SPE predictions, but Sadykov et al. [10] indicate that the timings of SPEs following flares can be unpredictable, stating that they can occur at any point up to 10 hours after their associated flares occur. To mitigate this concern, Sadykov et al. propose a forecast based on a combination of samples from ARs, soft X-ray (SXR) data, proton fluxes, and radio bursts. While this concept is promising, takes steps to resolve the aforementioned issue of vagueness in SPE forecasting in regard to the timings of SPEs following flares, and serves as a

good foundation for future works of a similar nature. there are flaws. Because each sample/record represents a full 24 hours of activity, precautions to mitigate the effects of SPEs would need to be in place for far longer than necessary if this system were to be implemented, unlike the solution presented in this work.

D. Improving Active Region-Based SPE Prediction

To help reduce the length of the prediction window, Kasapis et al. [7] propose a method by which the length of the prediction window is flexible, ranging from 2 to 10 hours. Like our work, the authors adopt SHMARP magnetic field parameters to make predictions. However, the authors match flares to ARs rather than using samples from ARs in forecasting, which means that SPEs caused by CMEs or events that are not yet known to us would remain undetected if this system were to be implemented.

E. Novelty and Contributions of Our Work

In contrast to the above studies, our work makes the following contributions.

- We collect samples/records of SHMARP magnetic field parameters or features from ARs and predict the association between the features and SPE occurrence. This new association prediction problem is motivated by the fact that photospheric magnetic field parameters from ARs can provide important information for SPEs that could possibly occur within the ARs. The temporal range of a sample/record is 96 minutes. This gives us results that allow for forecasts of SPEs that are not necessarily based on any events we know about without needing to wait for 24-hour cadences to elapse.
- This is the first study in which SHMARP magnetic field parameters are used to solve this association prediction problem, where SHMARP integrates SMARP magnetic parameters in Solar Cycle 23 (1996 - 2008) and SHARP magnetic parameters in Solar Cycles 24 (2008 - 2019) and 25 (2019 - present).
- We present a new framework (DeepSPE) in which we implement various sets of parameters for the RH algorithm to solve the problem at hand. We explore the hyperparameter space of RH in a comprehensive way by modifying the parameters presented by O’Keefe et al. [9], which has never been done before. Furthermore, DeepSPE is the first framework to solve the association prediction problem by integrating information from three solar cycles.

III. PROBLEM FORMULATION

DeepSPE deals with samples/records taken from ARs where each record comprises magnetic field parameter values in a 96-minute interval in an AR, where ARs on the Sun typically last between a few days and several weeks, with a median lifetime of roughly 5 to 8 days. Each record has a positive or negative label. A positive label indicates that an SPE occurs in the record’s 96-minute interval, while a negative label

TABLE I
MAGNETIC FIELD PARAMETERS AND THEIR DESCRIPTIONS

Feature	Description
R_VALUE	Unsigned flux near polarity inversion lines
USFLUXL	Line-of-sight unsigned flux
USFLUXZ	Vertical component of unsigned flux
MEANGBL_GMM	Mean of line-of-sight field gradient
MEANGBZ	Mean of vertical field gradient
CMASKL	Cylindrical equal area pixels in active region
RSUN_OBS	Observed angular radius of sun

indicates that no SPE occurs in the record's 96-minute interval. DeepSPE is trained using labeled training records. During testing, for a given test record whose label is absent, DeepSPE predicts its label. Thus, our model, in essence, predicts whether the magnetic field parameter values in the test record are associated with SPE occurrence.

IV. DATA PREPARATION

The SHMARP dataset, containing photospheric magnetic field parameters of solar active regions (ARs) in Solar Cycles 23-25, mainly surrounding sunspots, is used in this work. The SHMARP dataset is sampled in a mix of a 12-minute interval in Solar Cycles 24-25 and a 96-minute interval in Solar Cycle 23, which yields an initial set size of 742,125 records in the period between 1996 and 2022. To ensure consistency of scale across all features, all values in all columns (i.e., magnetic field parameters or features) containing negative number values are first shifted so that the lowest value in each column is zero. Then, normalization is performed by dividing the values in each column by its maximum value, ensuring that all values in each column range from zero to one. To harmonize the sampling frequency, the dataset is downsampled to averages within 96-minute sampling intervals, leaving 194,100 records to work with. Downsampling to a 96-minute interval was chosen over upsampling to a 12-minute interval due to the difficulty of predicting values of intermediate records that would be created within the 96-minute records collected during Solar Cycle 23. The seven columns (i.e., magnetic field parameters or features) used in our study are listed in Table I. The specific features listed here are chosen to eliminate redundancy within the dataset, eliminating situations whereby identical or very similar information may be provided by other features.

We employ the NOAA SPE Catalog [11] as a ground truth dataset to determine whether any record is associated with an SPE. The timestamps from this catalog are also downsampled to a 96-minute sampling interval and averaged, with the starting time being that of the SHMARP data, in order to ensure perfect matches between the two datasets' records, with any null records being deleted. An association between a given record having the seven magnetic field parameter values and an SPE is determined by whether the record's timestamp and unique AR number match those of any given SPE in the NOAA SPE Catalog. When there is a match, the record is labeled positive, indicating that the seven magnetic

field parameter values in the record's 96-minute interval are associated with an SPE. Otherwise, the record is labeled negative, indicating that the seven magnetic field parameter values in the record's 96-minute interval are not associated with an SPE, i.e., no SPE occurs in the 96-minute interval. This matching process yields 46 positive records and 194,054 negative records. Thus, our dataset is imbalanced.

To avoid very similar records existing in both the training and testing sets, records from 70% of all active regions (ARs) in our dataset are selected to be in the training set, with the remaining 30% of active regions (ARs) having records assigned to the testing set. Thus, the training set and the testing set are disjoint. Our model is tested in active regions (ARs) and records that it has never seen during training.

V. THE DEEPSPE MODEL

A. Model Architecture

DeepSPE employs the Random Hivemind V2 (RH v2) algorithm originating from [9] and alters various aspects of its architecture, treating them as hyperparameters, with the remaining aspects of the algorithm serving as a fixed base that is used throughout all tests. It works as an ensemble of individual neural networks, known as estimators, with varying input features, which can mitigate random weight and bias setups that perform poorly and give the ensemble as a whole more chances to make good predictions even if single members of the ensemble fail. In addition, the ensemble can be updated without the need to fully retrain it through the additions or removals of members. A "consensus" layer is added in order to decide the class (positive vs. negative) that the ensemble as a whole predicts, based on the outputs from each of its constituent members.

DeepSPE uses feature selection to minimize the possibility that redundant features are chosen in any given estimator. Each feature in Table I is first assigned an importance score. All scores are then normalized and act as the probabilities that each feature will be chosen in any estimator. Each estimator in DeepSPE is assigned an aggregate importance score equal to the sums of those of the features that have been chosen for it. These aggregate scores determine the values of estimators' epoch counts and learning rates. These aggregate scores also determine the amount of voting power each estimator has within an ensemble, as the output of the logistic layer of a DeepSPE estimator is multiplied by the sum of the estimator's feature scores; the class with the highest weighted average vote is the one the ensemble predicts. To determine these scores, the following formula is used:

$$s_i = \sqrt{\chi_i^2} + \kappa_i \quad (1)$$

Here, s_i represents the feature importance score of a given feature, χ_i^2 represents the chi square statistic between the feature and SPE appearance, and κ_i represents the mutual information gain statistic between the feature and SPE appearance.

TABLE II
CONFIGURATION DETAILS OF A DEEPSPE ESTIMATOR

Layer No.	Layer Type	Width/Dropout Probability
1	(Input)	4
2	Leaky Relu (0.01)	10
3	Leaky Relu (0.01)	10
4	Dropout	0.2
5	Linear	10
6	Softmax	2

The number of input features for each estimator is fixed at 4, as this is the value of half of the number of numerical features, rounded up. Each estimator in DeepSPE uses the Leaky ReLU and softmax activation function in hidden and output layers respectively. Leaky ReLU is chosen to avoid the possibility of a vanishing gradient caused by a zero-slope piece of the ReLU function. Two outputs are given by each estimator in the ensemble: one for the probability of membership of each class (i.e. positive vs. negative).

Each estimator in DeepSPE randomly selects exactly 4 features with the probability distribution described above. Each estimator is a deep neural network, whose configuration details are described in Table II. Table III summarizes the hyperparameters and their values used by an estimator, where Σs represents the sum of all estimators' feature importance ratings within a given ensemble.

Because the dataset at hand is imbalanced where negative samples significantly outnumber positive samples, we use a weighted cross entropy loss function to optimize the model parameters during training. The loss function is computed as:

$$J = \sum_{n=1}^N \sum_{k=1}^K \omega_k y_{nk} \log(\hat{y}_{nk}). \quad (2)$$

Here, N is the total number of records/samples in the training set, K is the number of classes, which is 2 in our case, since we only have positive and negative classes, ω_k is the weight of the k th class, which is derived by the ratio of the sizes of the positive and negative classes with more weight given to the minority (i.e., positive) class, and y_{nk} and $\hat{y}_{nk}$ denote the observed probability (which is equal to 1 if the n th record belongs to the k th class) and the estimated probability of being in the k th class of the n th record respectively. A weighted loss function is chosen over modifications to the dataset, as such modifications would have a chance, no matter how slight, of allowing for very few positive training records to exist within a possibly unrepresentative training sample.

Fig. 1 summarizes the architecture of a DeepSPE ensemble. The key in Fig. 1 describes the cell colors that are associated with each layer.

B. Boosting Schemes

Equations (3) through (5) define the three learning rate boosting schemes considered in this study, hereafter named "Legacy", "Modern", and "Experimental", represented by

TABLE III
ESTIMATOR TRAINING CONDITIONS AND HYPERPARAMETERS

Hyperparameter	Value
No. of Epochs	$100 \times (2 - \Sigma s)$
Base Learning Rate	0.001
Optimizer	Adam
Loss Function	Weighted cross entropy loss
Max. No. of Features	4

α_{Legacy} , α_{Modern} , and $\alpha_{\text{Experimental}}$, respectively. To determine the boosted learning rate of an estimator, the base learning rate (α_0) is multiplied by the multiplier corresponding to that of the given scheme being tested.

$$\alpha_{\text{Legacy}} = \alpha_0 \times \frac{es(\Sigma s + M_{\text{input}})}{M_{\text{input}}} \quad (3)$$

$$\alpha_{\text{Modern}} = \alpha_0 \times (0.5 + s) \quad (4)$$

$$\alpha_{\text{Experimental}} = \alpha_0 \times \frac{s(\Sigma s + M_{\text{input}})}{M_{\text{input}}} \quad (5)$$

In Table III and Equations 3 through 5, e represents the base of the natural logarithm, s represents the sum of a given estimator's feature importance ratings, Σs represents the sum of all estimators' feature importance ratings within a given ensemble, and M_{input} represents the number of features passed to a given estimator as input.

C. Prediction Methods

In this work, two individual estimator prediction methods are compared. "Strict" estimator prediction employs a "winner take all" system whereby the class that is most likely to occur receives all of a given estimator's vote and the other class receives nothing from that estimator. "Probabilistic" estimator prediction employs a "weighted" system whereby the outputs of the logistic layers of each estimator are summed together.

VI. EXPERIMENTS AND RESULTS

A. Experimental Setup

We define a sample/record as a true positive if our DeepSPE model predicts that the record is positive, and the record is indeed positive, i.e., the magnetic field parameters in the record are associated with an SPE. We define a record as a false positive if our model predicts that the record is positive while the record is actually negative, i.e., the magnetic field parameters in the record are not associated with an SPE. We say a record is a true negative if our model predicts the record to be negative and the record is indeed negative; a record is a false negative if our model predicts the record to be negative while the record is actually positive. We use TP, FP, TN, and FN to represent the total number of true positives, false positives, true negatives, and false negatives respectively from each series of tests.

In addition to accuracy (ACC), balanced accuracy (BACC) and true skill statistic (TSS) scores are employed to offset the effect of class imbalance on ACC.

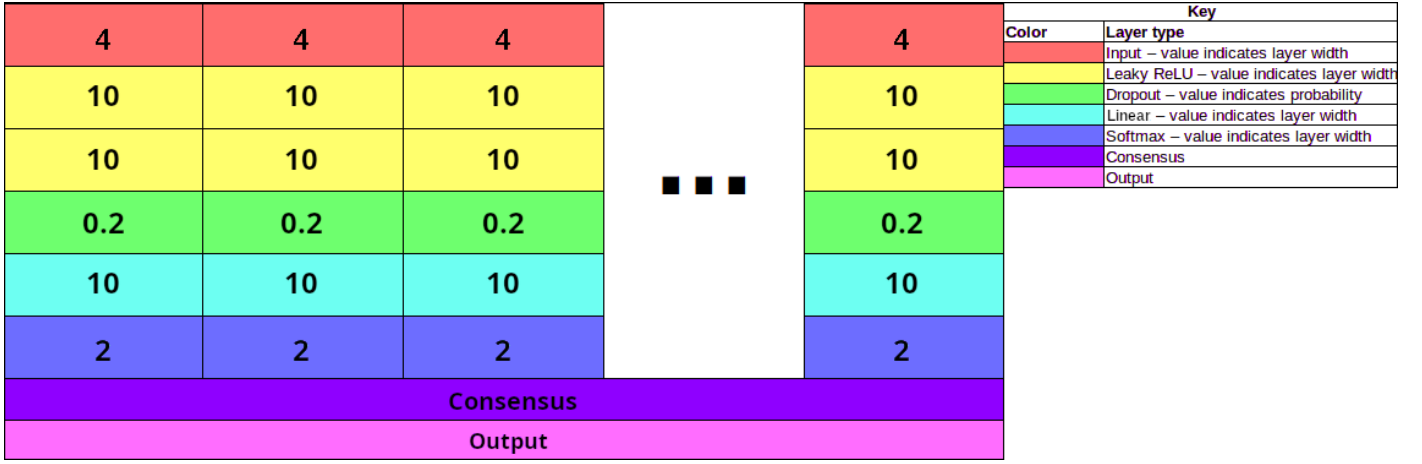


Fig. 1. DeepSPE Ensemble Architecture

The performance metrics used in our study include the following:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}, \quad (6)$$

$$BACC = \frac{TP(TN + FP) + TN(TP + FN)}{2(TN + FP)(TP + FN)}, \quad (7)$$

$$TSS = \frac{TP}{TP + FN} - \frac{FP}{TN + FP}. \quad (8)$$

ACC and BACC range between 0 and 1, with higher values indicating better performance. TSS ranges between -1 and 1 , which compares the true positive rate (first term in Equation (8)) with the false positive rate (second term in Equation (8)). TSS = 0 indicates an exact balance between the true positive rate and the false positive rate. In addition, we use ROC_AUC, which serves as an indicator of robustness [9]. To calculate ROC_AUC, the prediction probabilities of each class for a DeepSPE ensemble are determined by multiplying those of each of its estimators by each of their feature score sums following normalization and using the weighted averages of the probabilities associated with each class among all estimators in a given ensemble. The formula for this is given by Equation 9, where $P(C = \alpha)$ represents the probability of a prediction for a given class given a set of feature values as input, $n_{estimators}$ represents the number of estimators in a given ensemble, $x_{\alpha,i}$ represents the output of a given estimator's logistic layer for a given class, s_i represents the sum of a given estimator's column weights, and $\sum s$ represents the sum of all estimators' column weights across the entire ensemble.

$$P(C = \alpha) = \sum_{i=1}^{n_{estimators}} \frac{x_{\alpha,i} s_i}{\sum s} \quad (9)$$

In our experiments, we also compare DeepSPE with a baseline method, which is a SiNN (single neural network) without any boosting, ensemble voting, or feature selection, acting alone instead of as part of an ensemble. The architecture of this SiNN is identical to those of the estimators used in the other tests, but with all 7 of the input features used instead of

just a random selection of 4. The remaining hyperparameters are also the same, except the number of epochs is fixed at 100 and the learning rate is fixed at 0.001. The same dataset is used to train and test the baseline method with the same method of separating the data into the training and testing sets.

B. Experimental Results

Each of the three learning rate boosted schemes is tested using ensembles containing 10, 15, 20, 25, 30, 40, 60, and 100 estimators, respectively. All models are run 50 times and are evaluated using the metrics defined in Equations (6) through (8). The averages and standard deviations of each performance metric from the top 5 results by TSS of the 50 tests of each of the learning rate boosting schemes and numbers of estimators per ensemble tested are used to evaluate each configuration. Table IV shows the best and worst results for each metric for all DeepSPE tests that fit the prior criterion. Each of these results from the various “winners” and “losers” of each metric are displayed alongside whether each test was the best or worst test of its given metric along with its number of estimators, boosting scheme, and prediction method. Table V shows the results for the baseline test.

The model that performed the best in terms of both balanced accuracy and TSS had Accuracy and ROC_AUC scores in line with the other configurations tested; no configuration had a difference greater than 10% in the averages of either metric compared to it. All models tested here seem to exhibit a high degree of robustness; the ROC_AUC serves as an indicator of robustness [9] and the lowest ROC_AUC score obtained from any of the deep learning tests was 0.8516.

The baseline SiNN performed close to the lower bound of the DeepSPE models by balanced accuracy, TSS score, and ROC_AUC. The worst average balanced accuracy score from a DeepSPE model was 0.7951 and that from the SiNN was 0.8060. The worst average TSS score from a DeepSPE model was 0.5903 and that from the SiNN was 0.6121. The worst average ROC_AUC from a DeepSPE model was 0.8516 and that from the SiNN was 0.8581.

TABLE IV
BEST AND WORST RESULTS FOR EACH PERFORMANCE METRIC

Performance Metric	Best/Worst	No. of Estimators	Boosting Scheme	Prediction Method	Average
ACC	Best	20	Modern	Probabilistic	0.7724 ± 0.0623
	Worst	20	Legacy	Probabilistic	0.6558 ± 0.0379
BACC	Best	30	Modern	Probabilistic	0.8358 ± 0.0171
	Worst	100	Experimental	Strict	0.7951 ± 0.0140
TSS	Best	30	Modern	Probabilistic	0.6716 ± 0.0342
	Worst	100	Experimental	Strict	0.5903 ± 0.0279
ROC_AUC	Best	100	Legacy	Probabilistic	0.9077 ± 0.0147
	Worst	25	Experimental	Strict	0.8516 ± 0.0244

TABLE V
SINGLE NEURAL NETWORK RESULTS

Performance Metric	Average
ACC	0.7707 ± 0.0365
BACC	0.8060 ± 0.0365
TSS	0.6121 ± 0.0729
ROC_AUC	0.8581 ± 0.0527

VII. DISCUSSION AND CONCLUSION

In this work, we introduced a novel ensemble deep learning framework, DeepSPE, and evaluated how well various sets of model parameters in the framework performed. The model parameters include the number of estimators in a given model, the manner in which the learning rates of the estimators are boosted or attenuated, and the manner in which individual estimators formulate their predictions. The key results of this work are:

- A DeepSPE ensemble model with more estimators does not always guarantee better results. The highest TSS score of any test, 0.6716, was obtained with an ensemble of 30 estimators despite the fact that other ensembles were tested using higher numbers of estimators.
- The Modern learning rate boosting scheme outperformed the Legacy and Experimental schemes, which only made learning rates faster. 7 of the top 12 and 13 of the top 24 TSS scores were obtained using the Modern learning rate boosting scheme.
- The Probabilistic class prediction method, which gave weighted votes to classes based on their respective probabilities of occurrence, outperformed the Strict method, which only gave votes to the classes that were the most likely to occur. 7 of the top 12 and 15 of the top 24 TSS scores were obtained using the Probabilistic class prediction method.
- All of the models seem to exhibit a great degree of robustness; the lowest ROC_AUC score obtained from any of the deep learning tests was 0.8516.
- The baseline SiNN performed close to the worst ensemble model by balanced accuracy, TSS score, and ROC_AUC. The worst average balanced accuracy score from an ensemble model was 0.7951 and that from the SiNN was 0.8060. The worst average TSS score from an ensemble model was 0.5903 and that from the SiNN was 0.6121. The worst average ROC_AUC from an ensemble

model was 0.8516 and that from the SiNN was 0.8581. These scores represent a slight improvement over the worst performing ensemble model tested by these metrics, but given that there were also several other ensemble models that performed better by these same metrics, the result presented here does not establish the SiNN as a comparable model.

From these results, we conclude that DeepSPE is a valid framework that can predict the association between magnetic field features and SPE occurrence, regardless of class imbalance.

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